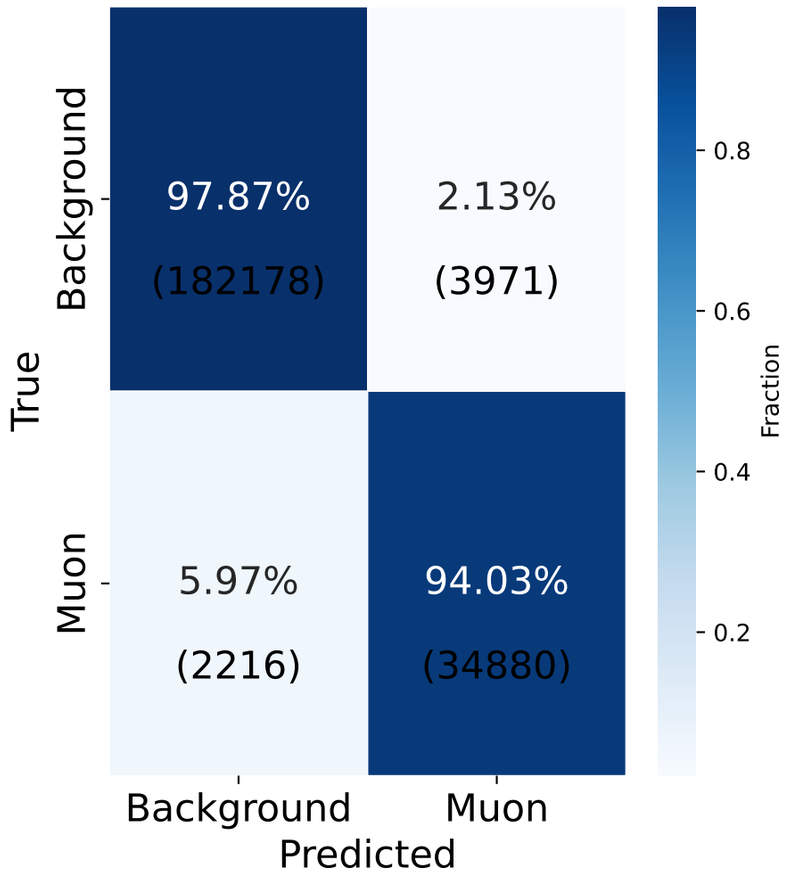
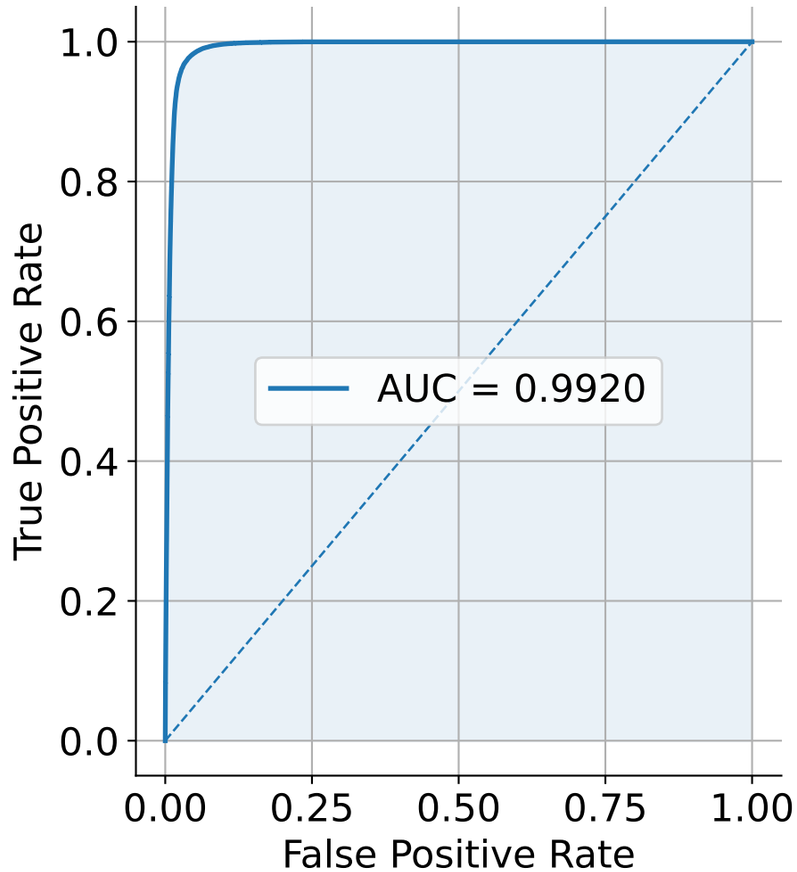
Muon Identification — XGBoost Analysis (hit-based features)
[Train S/B=1:5 | Test S/B=1:5 (realistic, all files)]

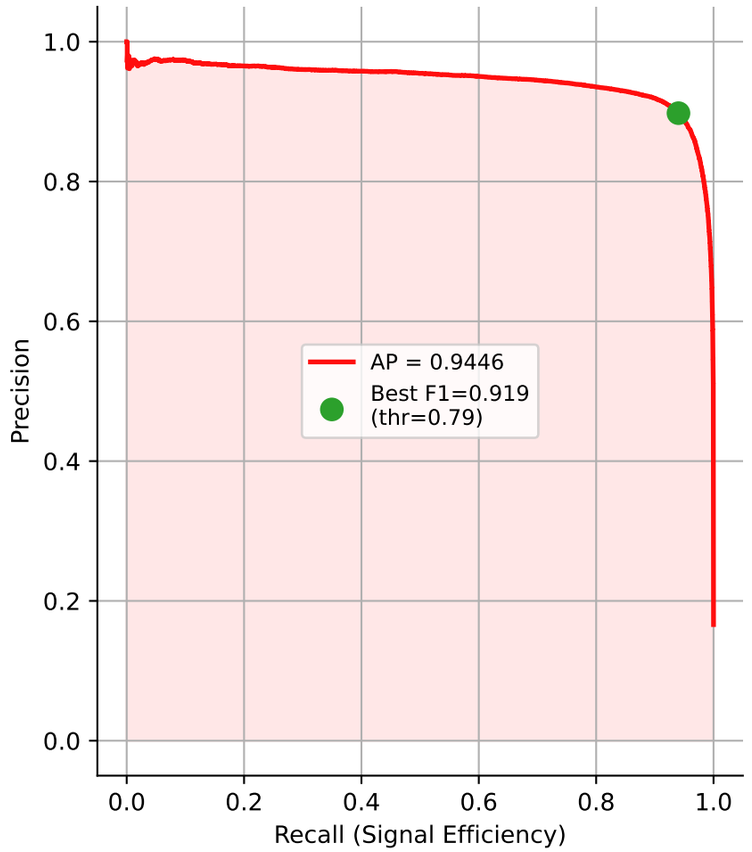
Confusion Matrix @ Max F1-Score (thr=0.794)



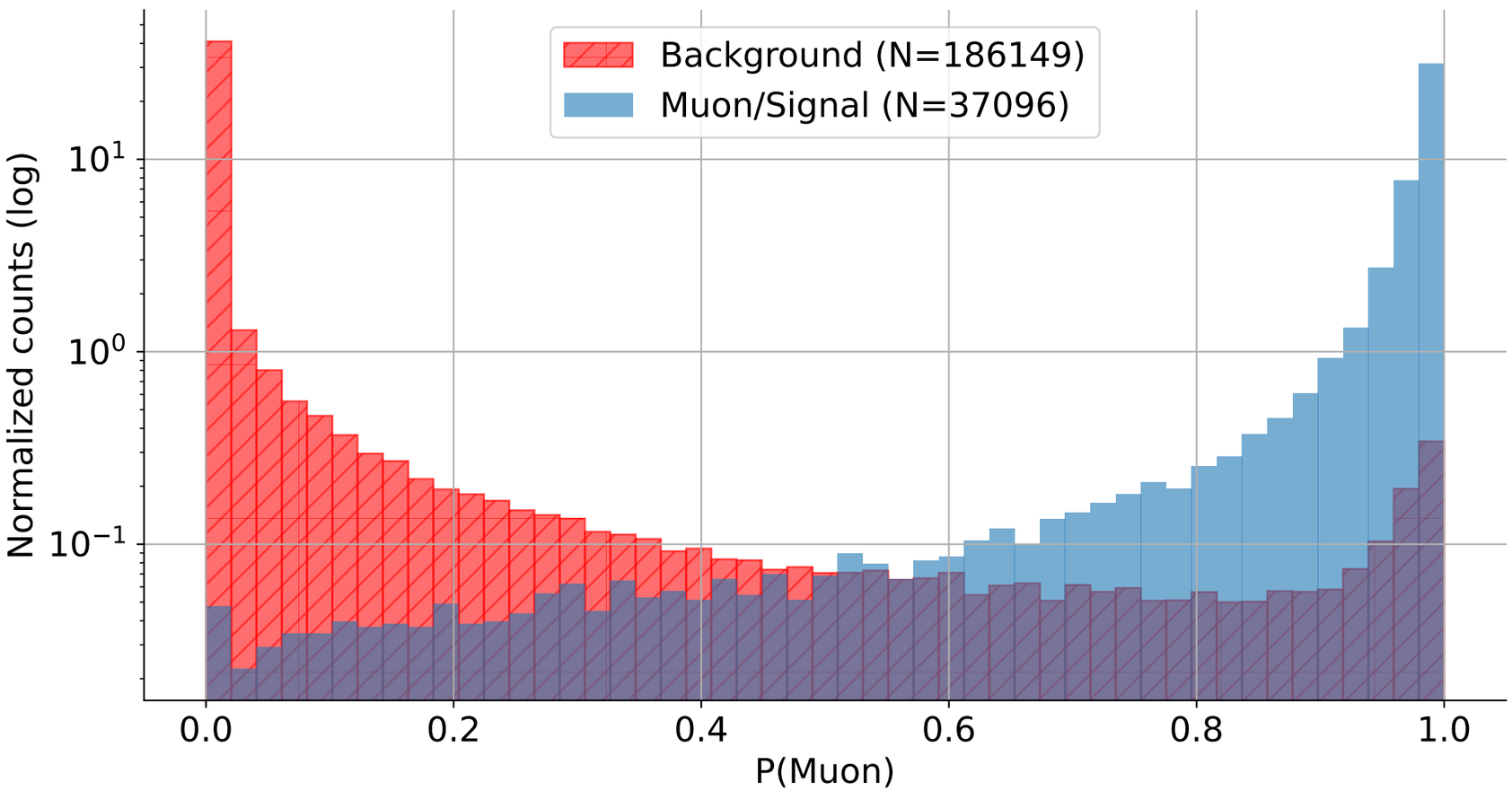
ROC Curve



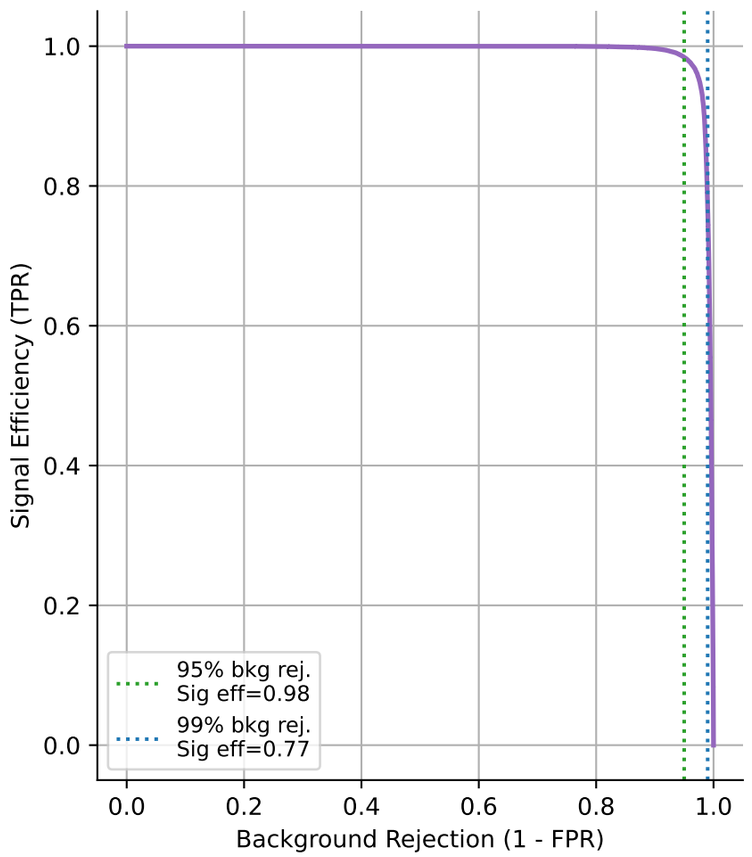
Precision-Recall Curve



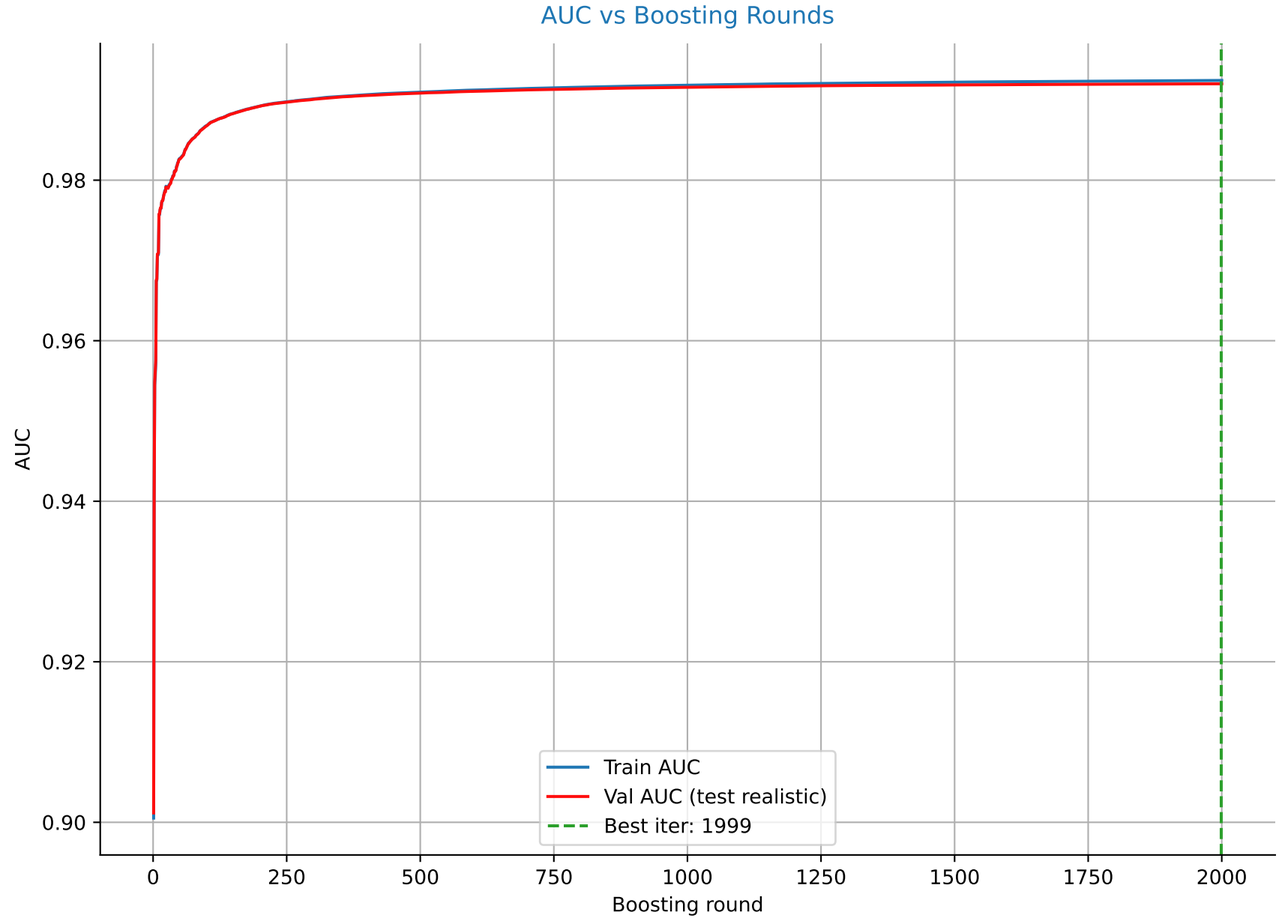
Classifier Response Distribution [Test: realistic S/B]

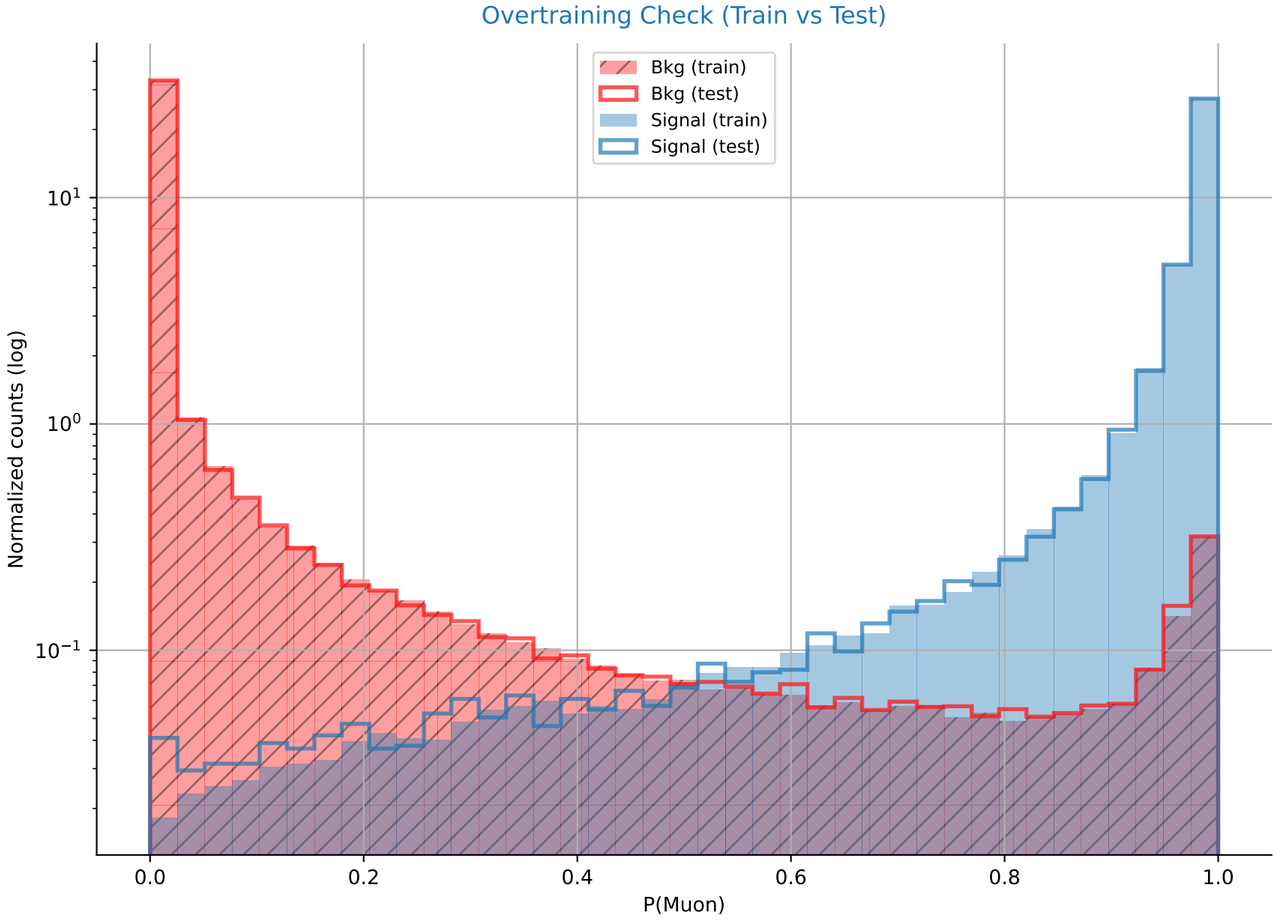


Signal Eff. vs Background Rejection



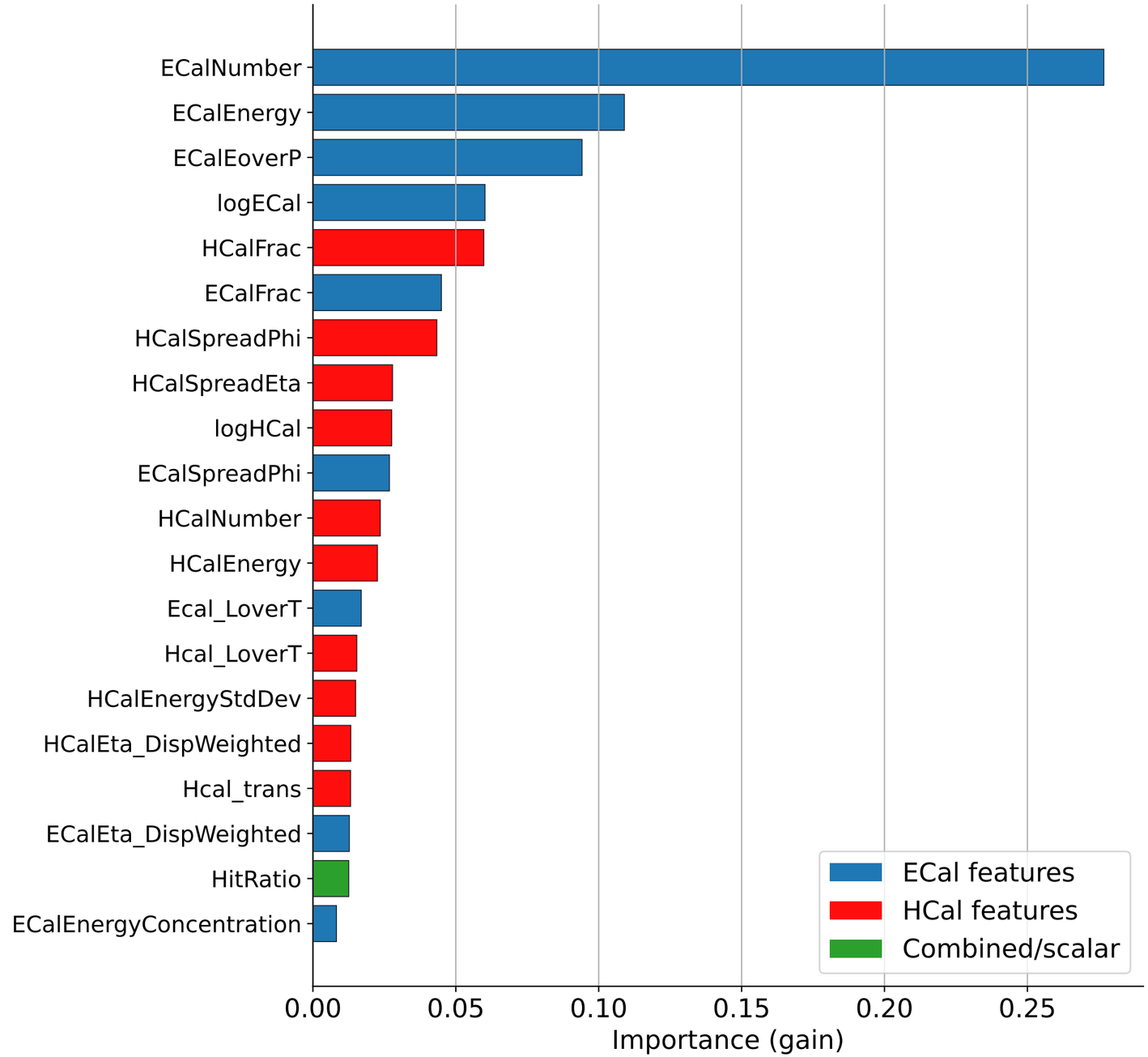
XGBoost Training Diagnostics



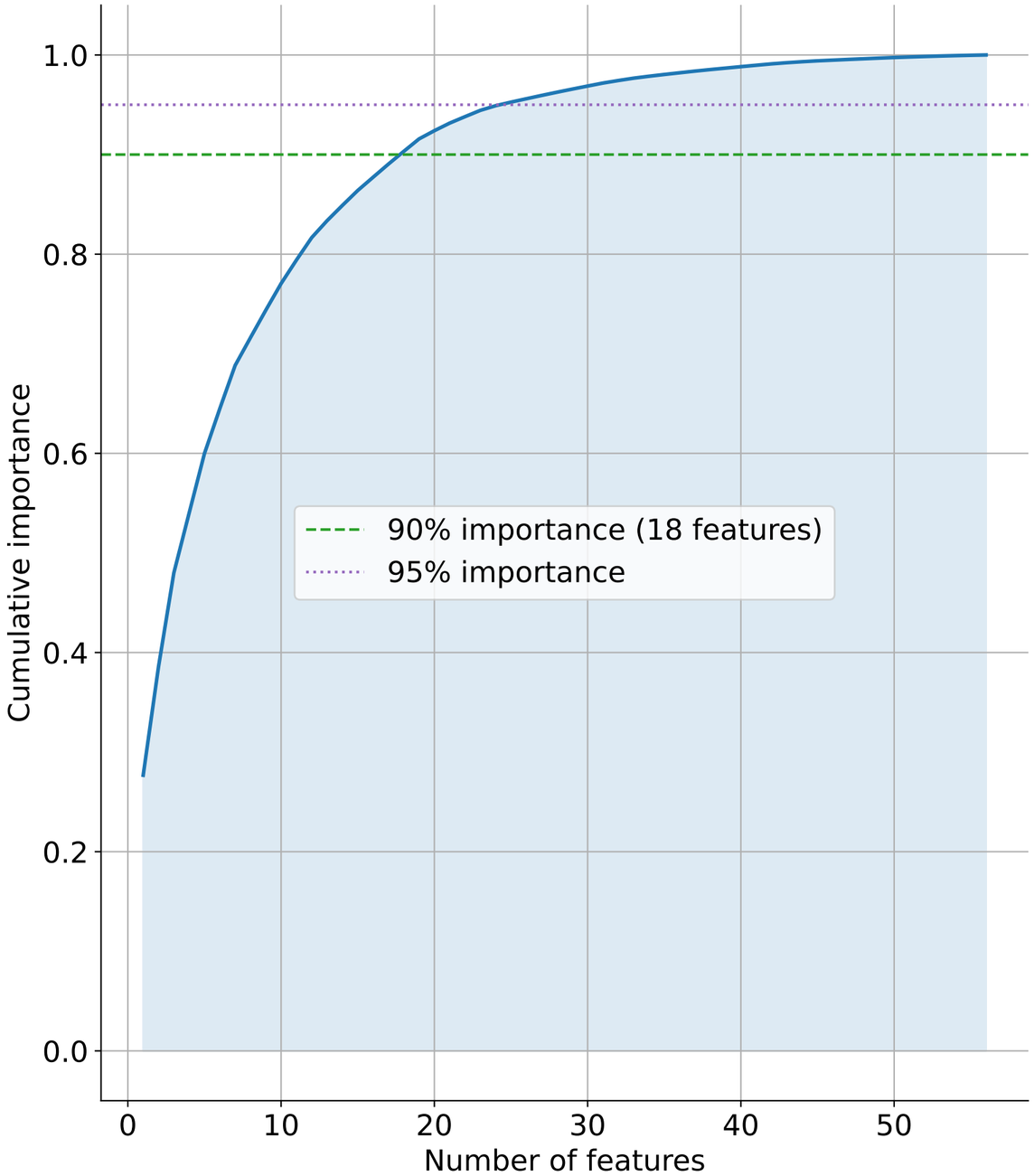


Feature Importances — XGBoost

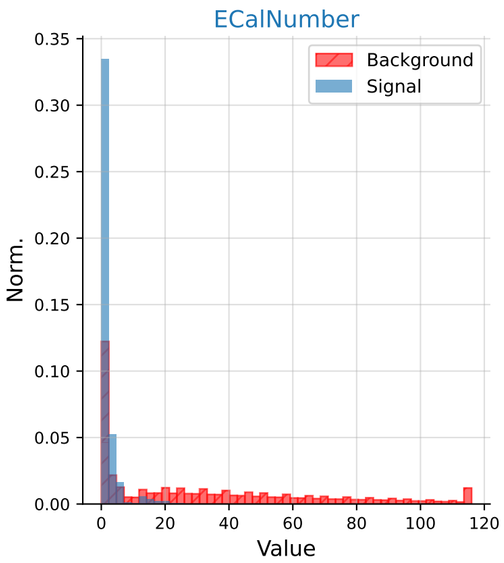
Top 20 Features

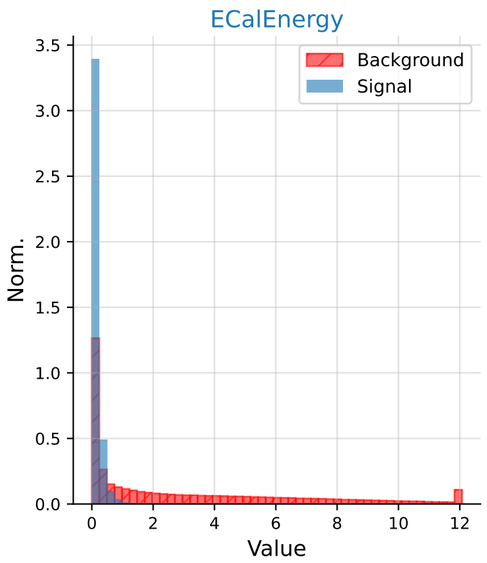


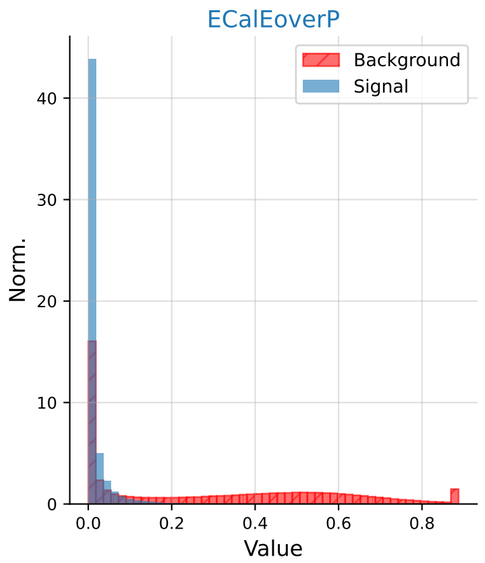
Cumulative Feature Importance

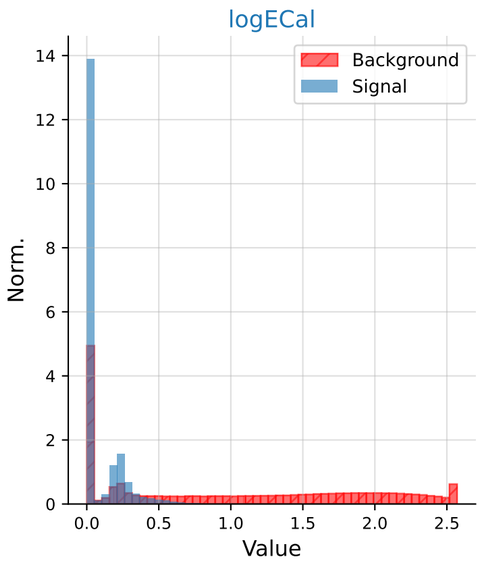


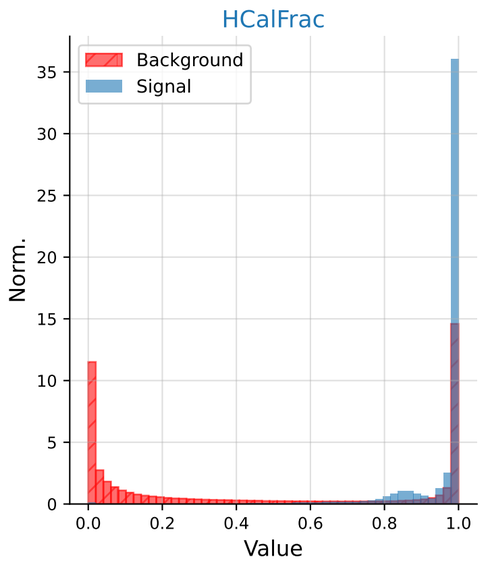
Key Physics Variable Distributions: Signal vs Background
[full dataset before splitting]

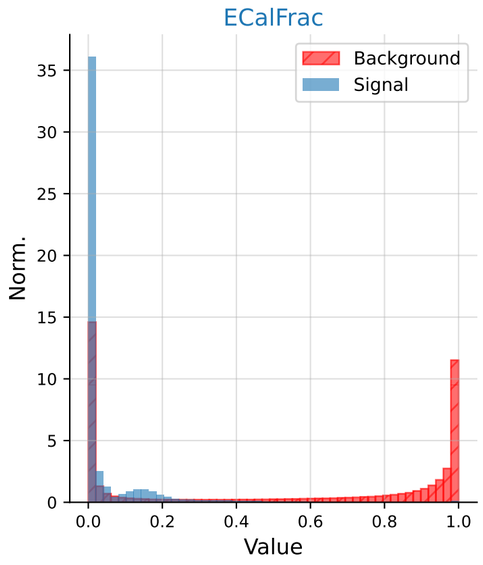


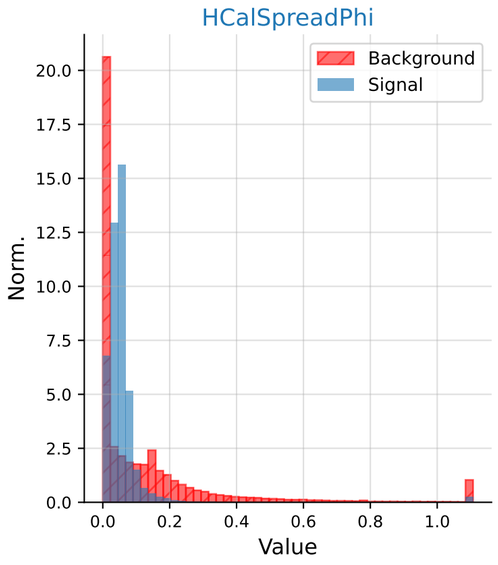


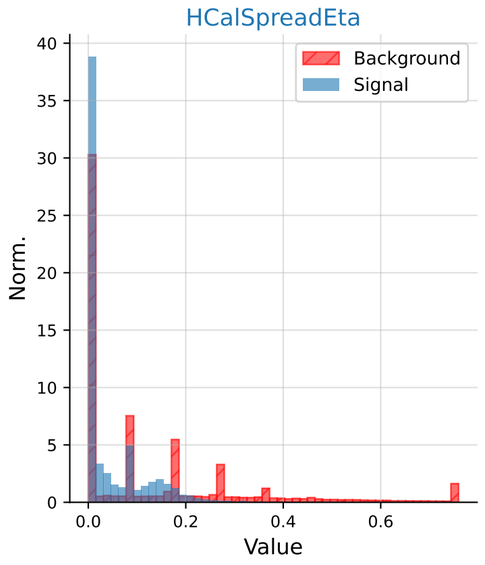


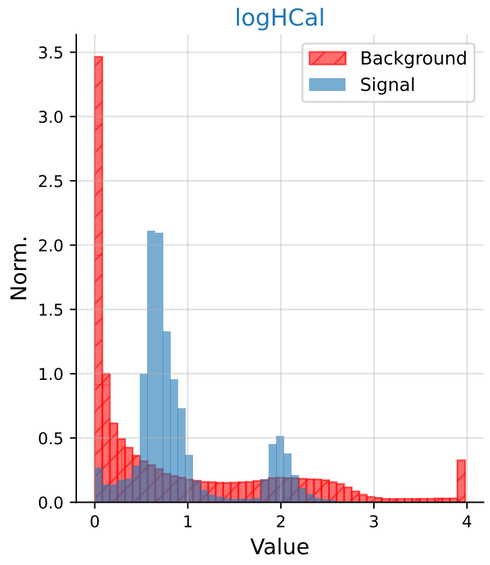


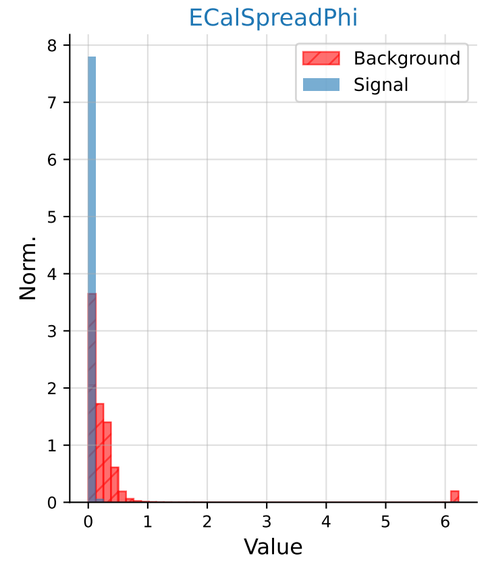


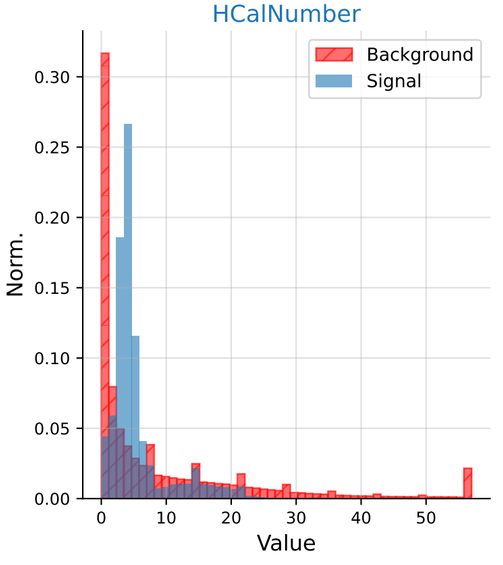


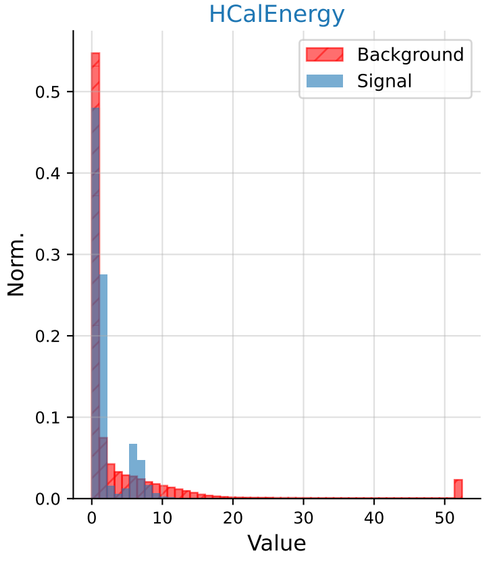


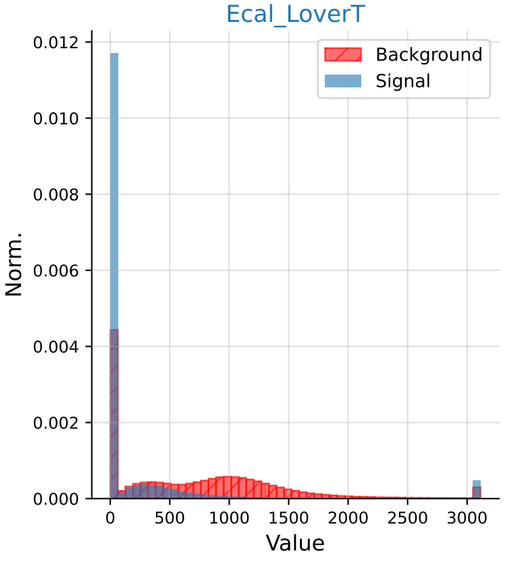


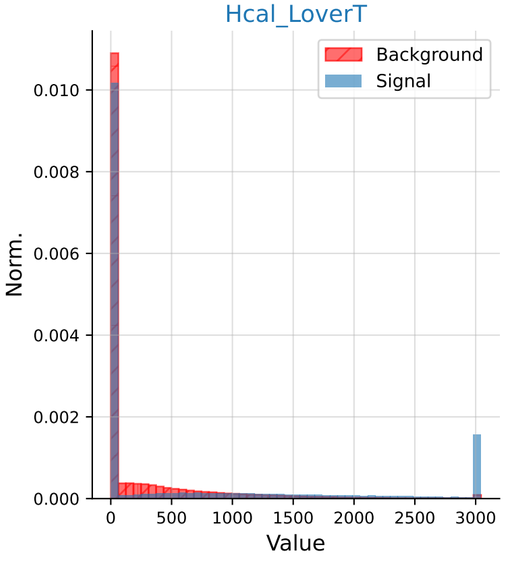


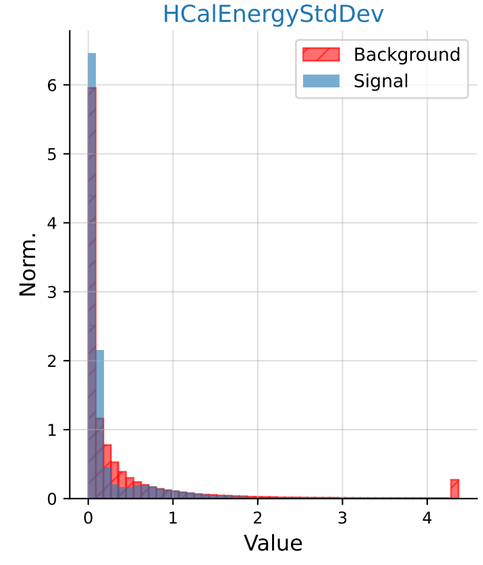






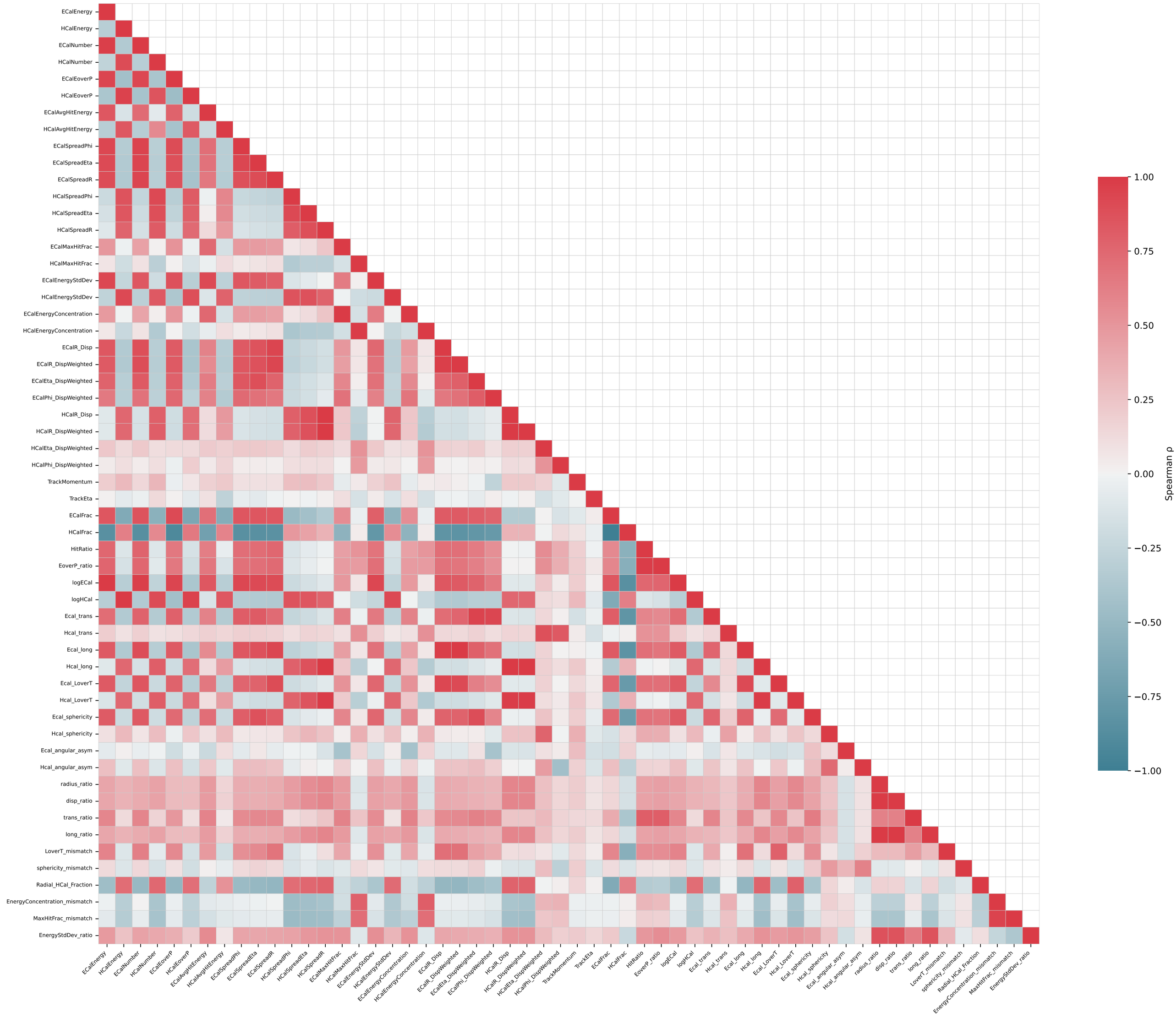






Spearman Correlation Matrix — All Features
(values from the full dataset)

Higher |ρ| means a stronger monotonic relationship between features



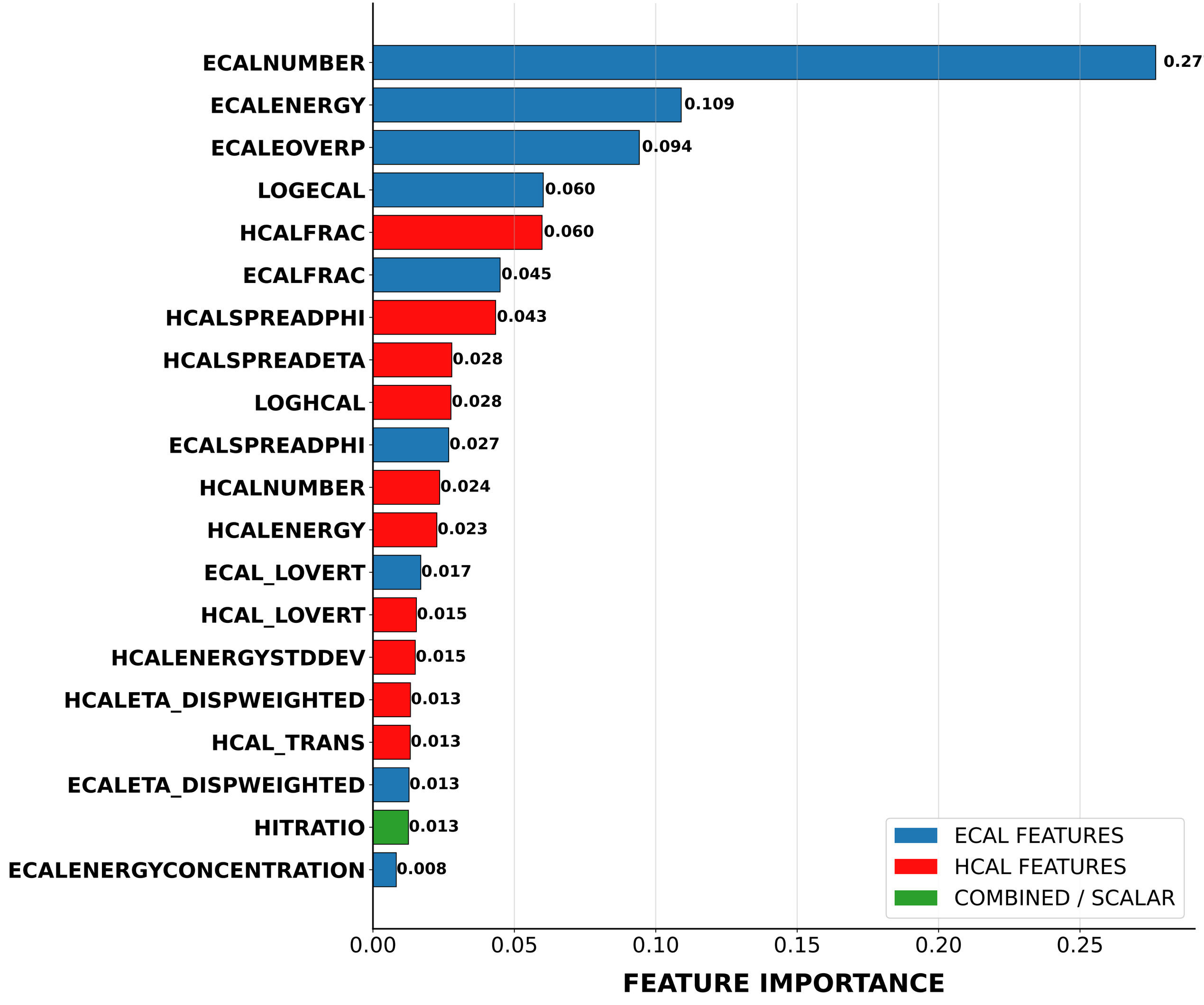
Feature correlation with signal/background class
Blue = positive (→ muon), Red = negative (→ background)

TrackMomentum
TrackEta
Ecal_angular_asym
EnergyConcentration_mismatch
HCalSpreadPhi
HCalSpreadR
HCalSpreadEta
HCalR_Disp
HCalEnergyConcentration
MaxHitFrac_mismatch
Hcal_long
HCalR_DispWeighted
HCalNumber
HCalEnergyStdDev
Hcal_LoverT
HCalMaxHitFrac
HCalPhi_DispWeighted
HCalEnergy
logHCal
HCalEoverP
radius_ratio
long_ratio
disp_ratio
EnergyStdDev_ratio
Hcal_sphericity
trans_ratio
Radial_HCal_Fraction
LoverT_mismatch
ECalEnergyConcentration
HCalAvgHitEnergy
ECalMaxHitFrac
sphericity_mismatch
Hcal_trans
HCalEta_DispWeighted
HitRatio
Hcal_angular_asym
EoverP_ratio
ECalAvgHitEnergy
ECalPhi_DispWeighted
Ecal_LoverT
Ecal_sphericity
Ecal_trans
ECalR_DispWeighted
Ecal_long
ECalR_Disp
ECalEnergyStdDev
ECalEta_DispWeighted
ECalFrac
HCalFrac
ECalSpreadR
ECalSpreadEta
ECalSpreadPhi
logECal
ECalEnergy
ECalEoverP
ECalNumber

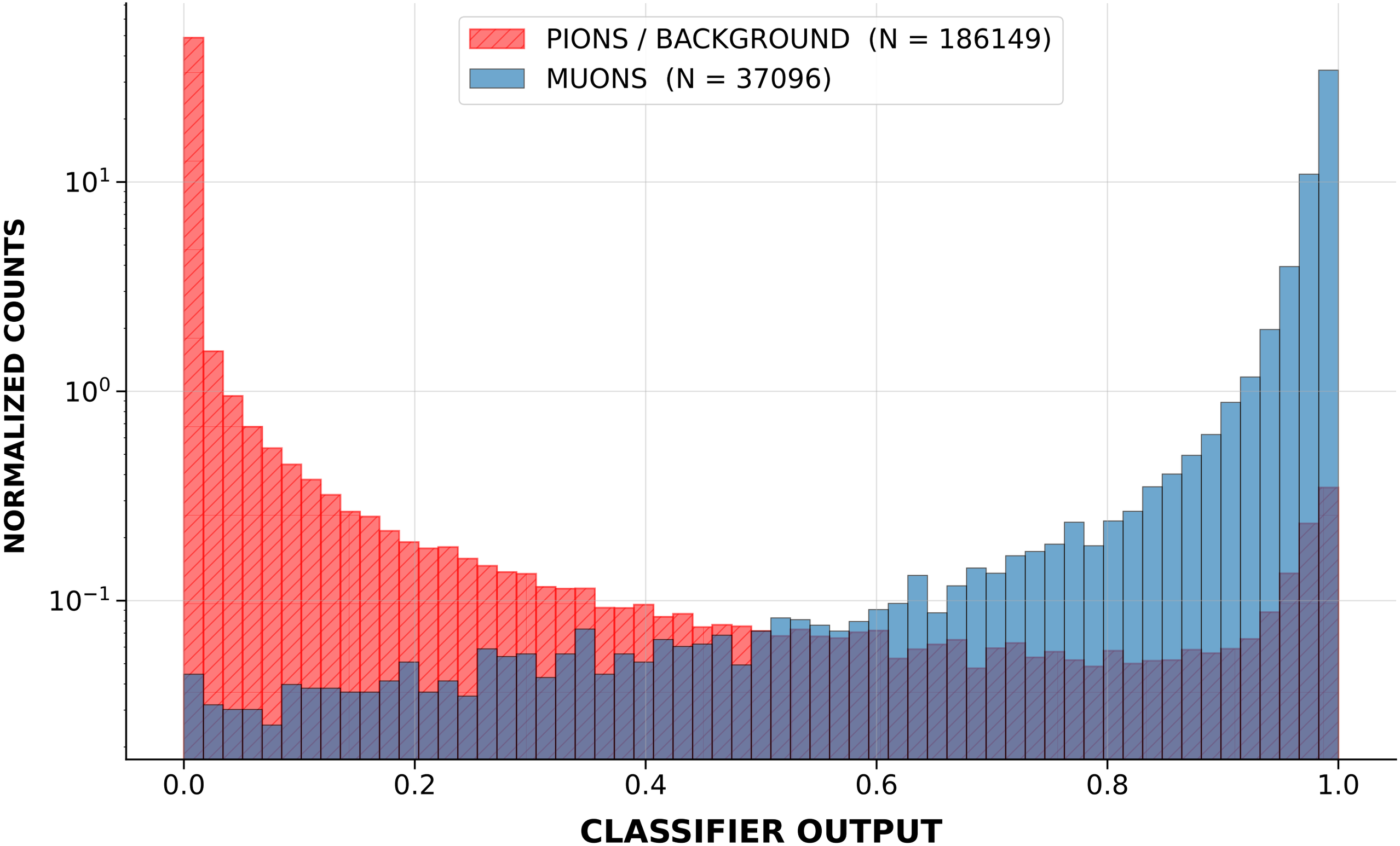
−0.4 −0.3 −0.2 −0.1 0.0 0.1 0.2 0.3 0.4

Spearman ρ with the label (IsMuon)

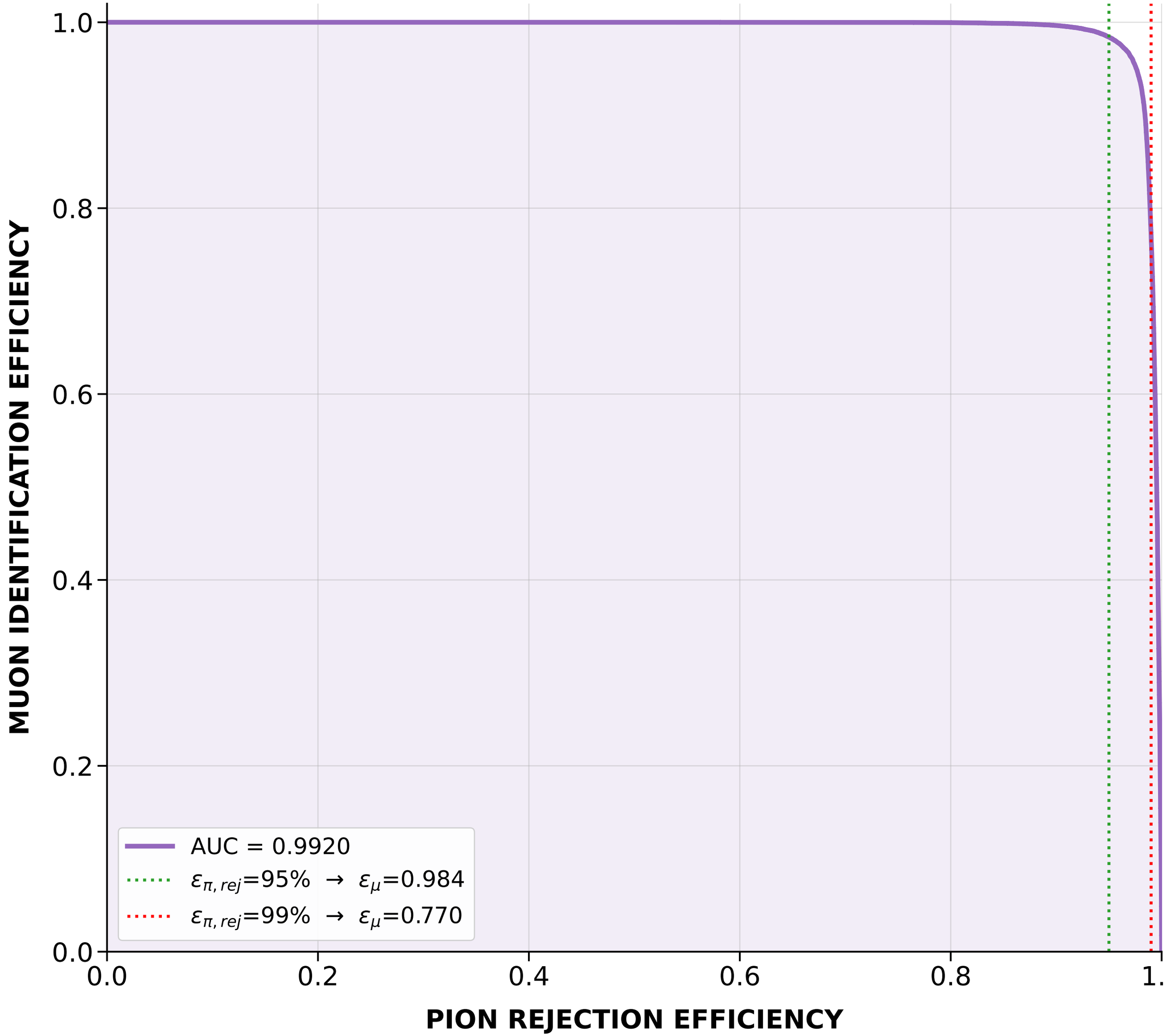
FEATURE IMPORTANCE — TOP 20



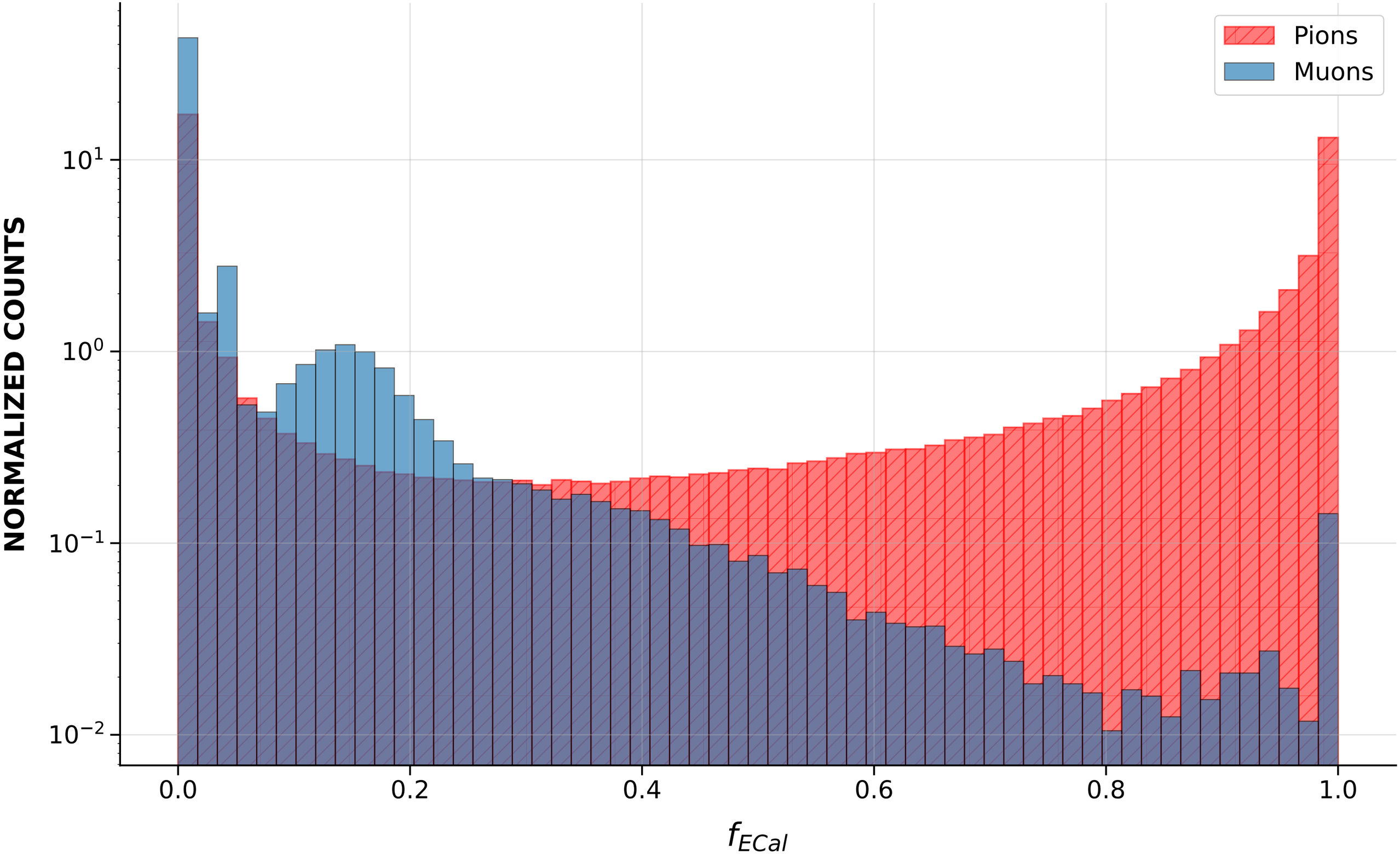
CLASSIFIER SCORE DISTRIBUTION
MUONS vs PIONS



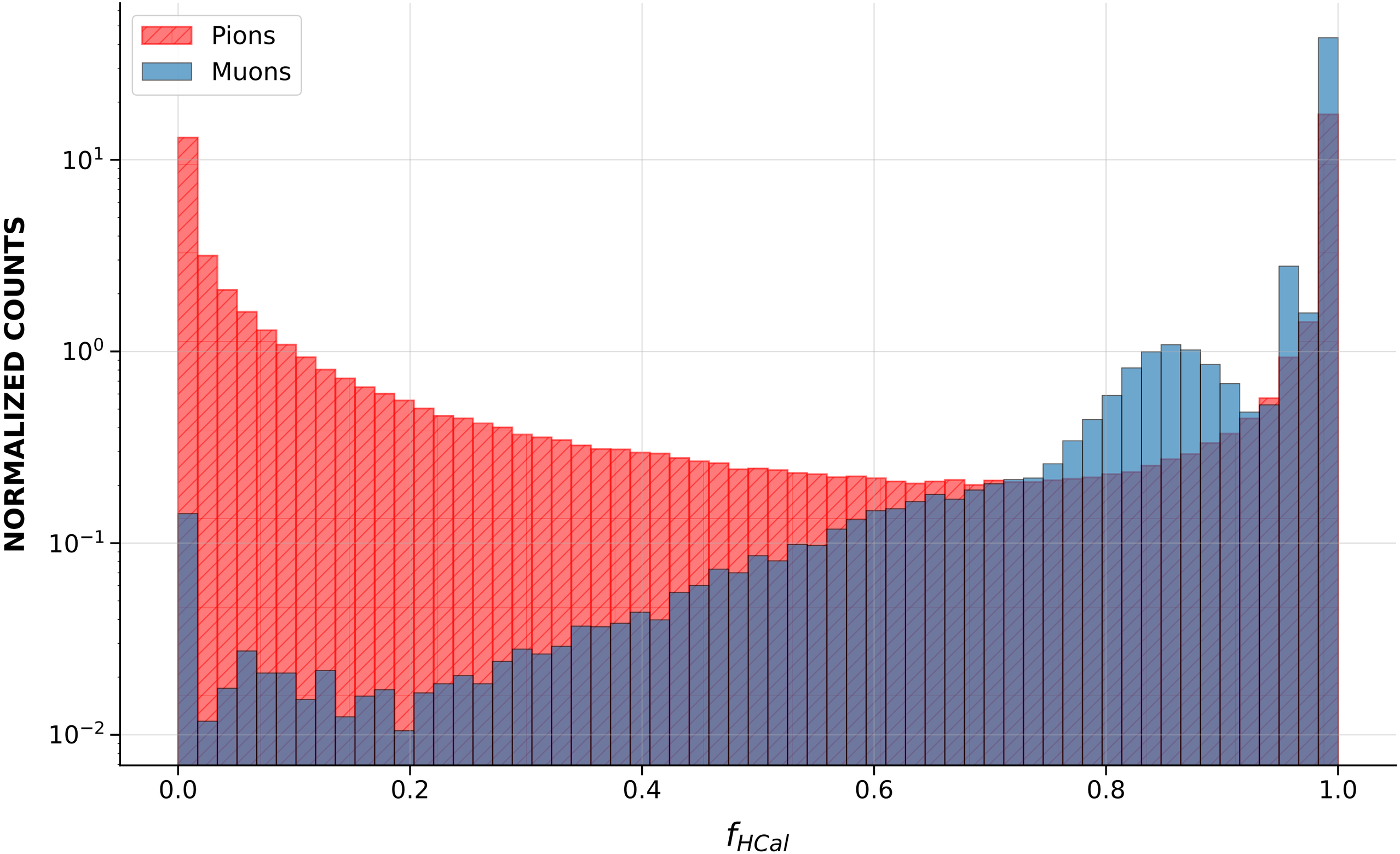
ROC CURVE: ε_μ vs $\varepsilon_{\pi,\text{rej}}$



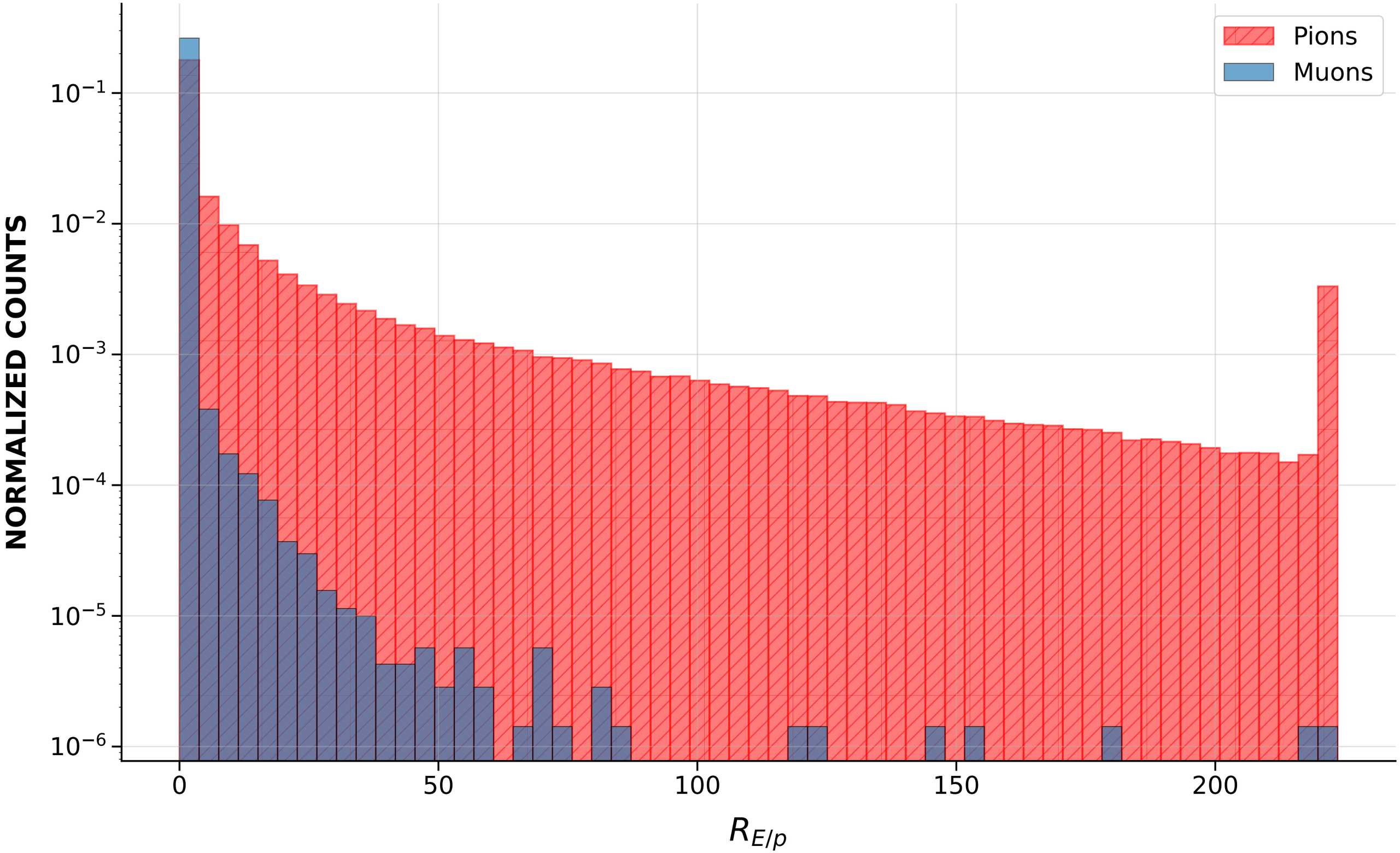
ENGINEERED CALORIMETER FEATURE
ECalFrac



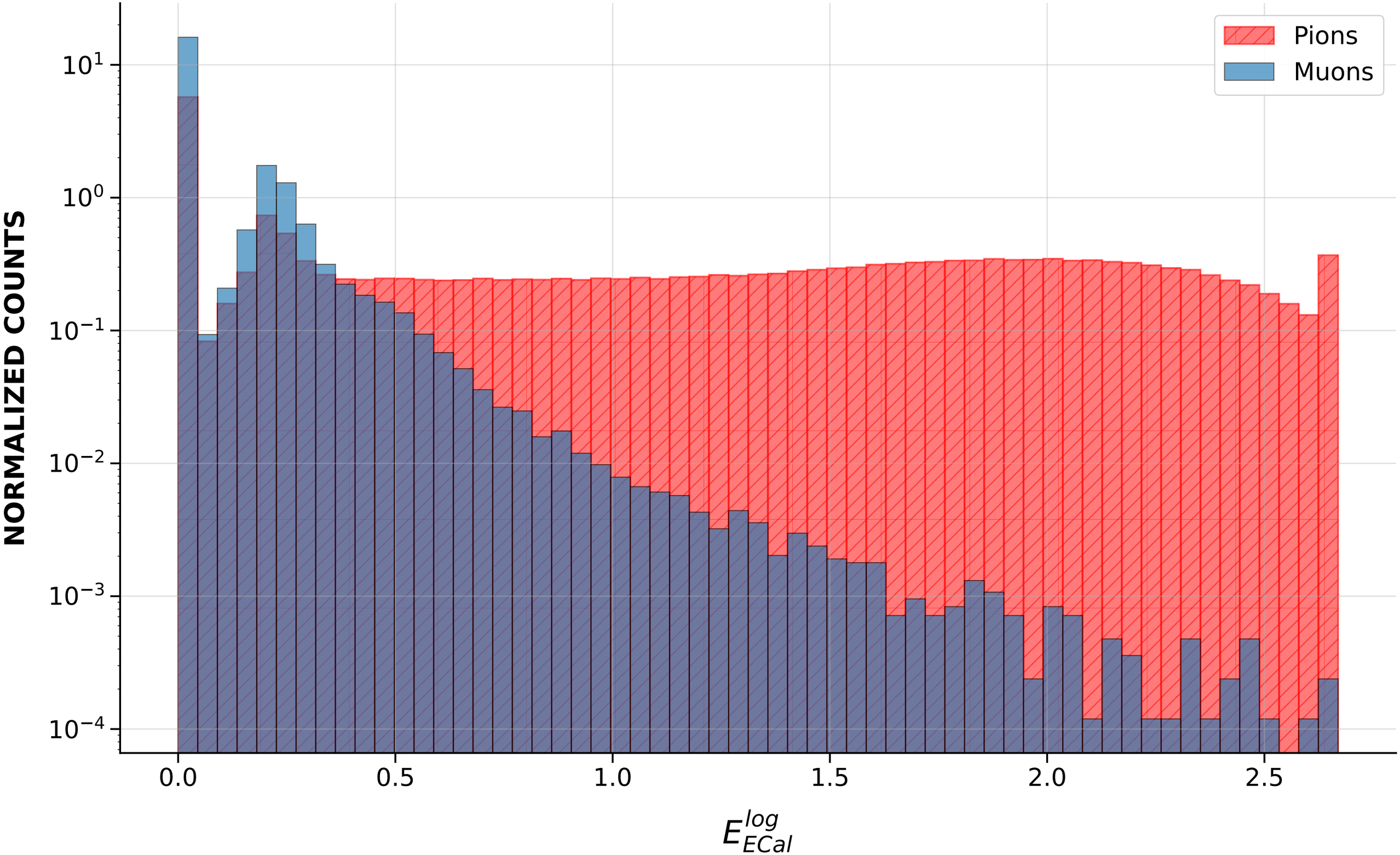
ENGINEERED CALORIMETER FEATURE
HCalFrac



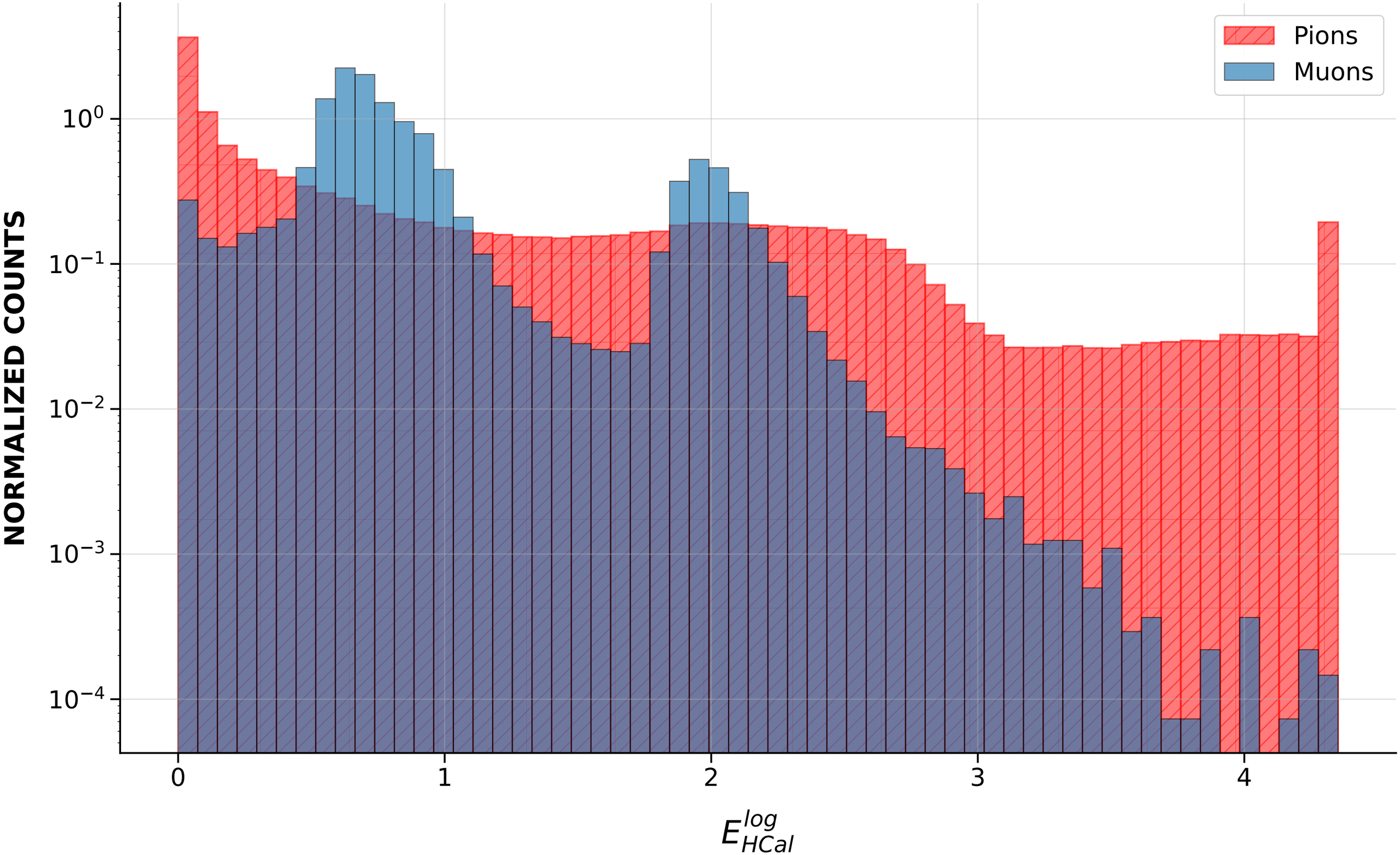
ENGINEERED CALORIMETER FEATURE
EoverP_ratio



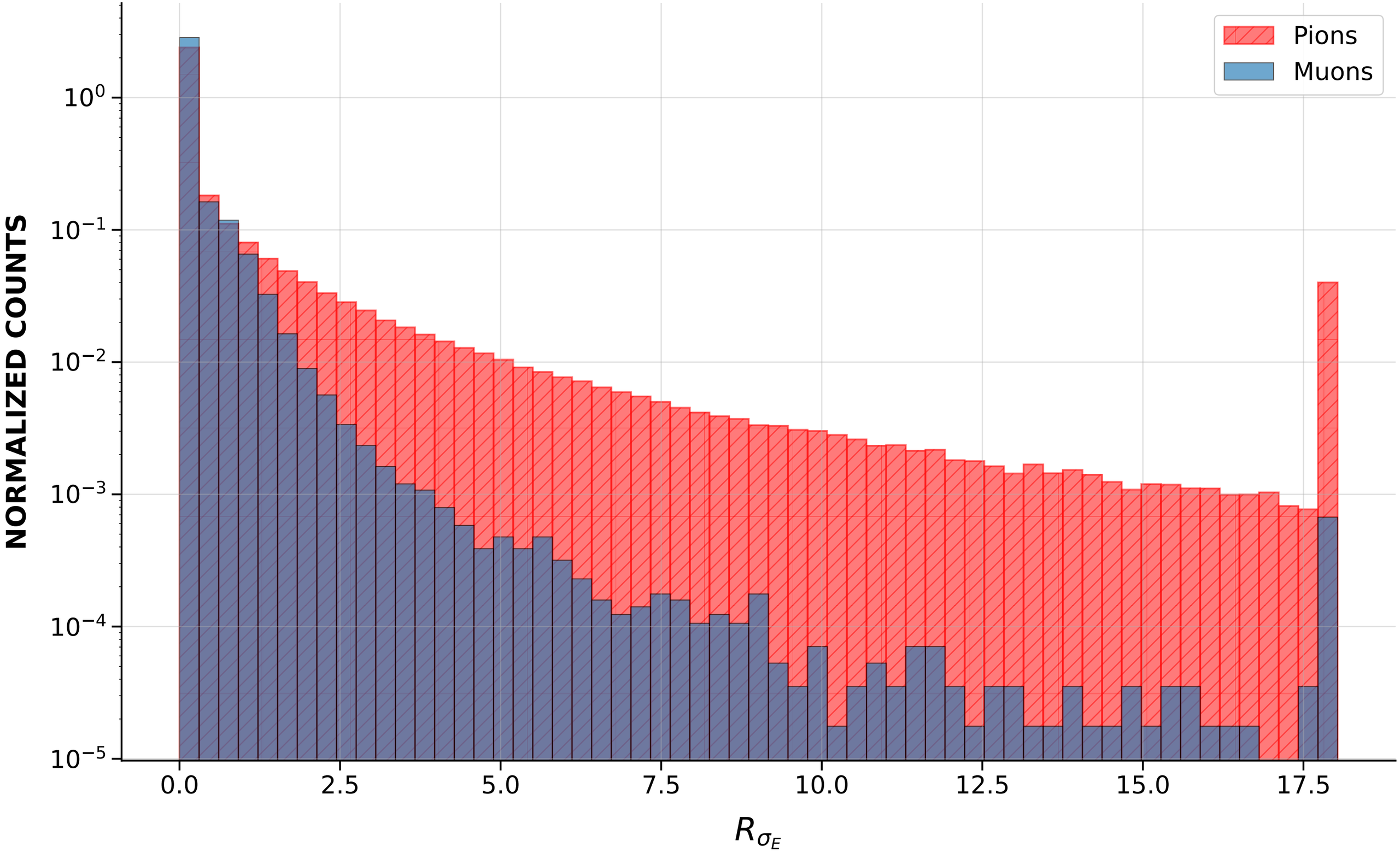
ENGINEERED CALORIMETER FEATURE
logECal



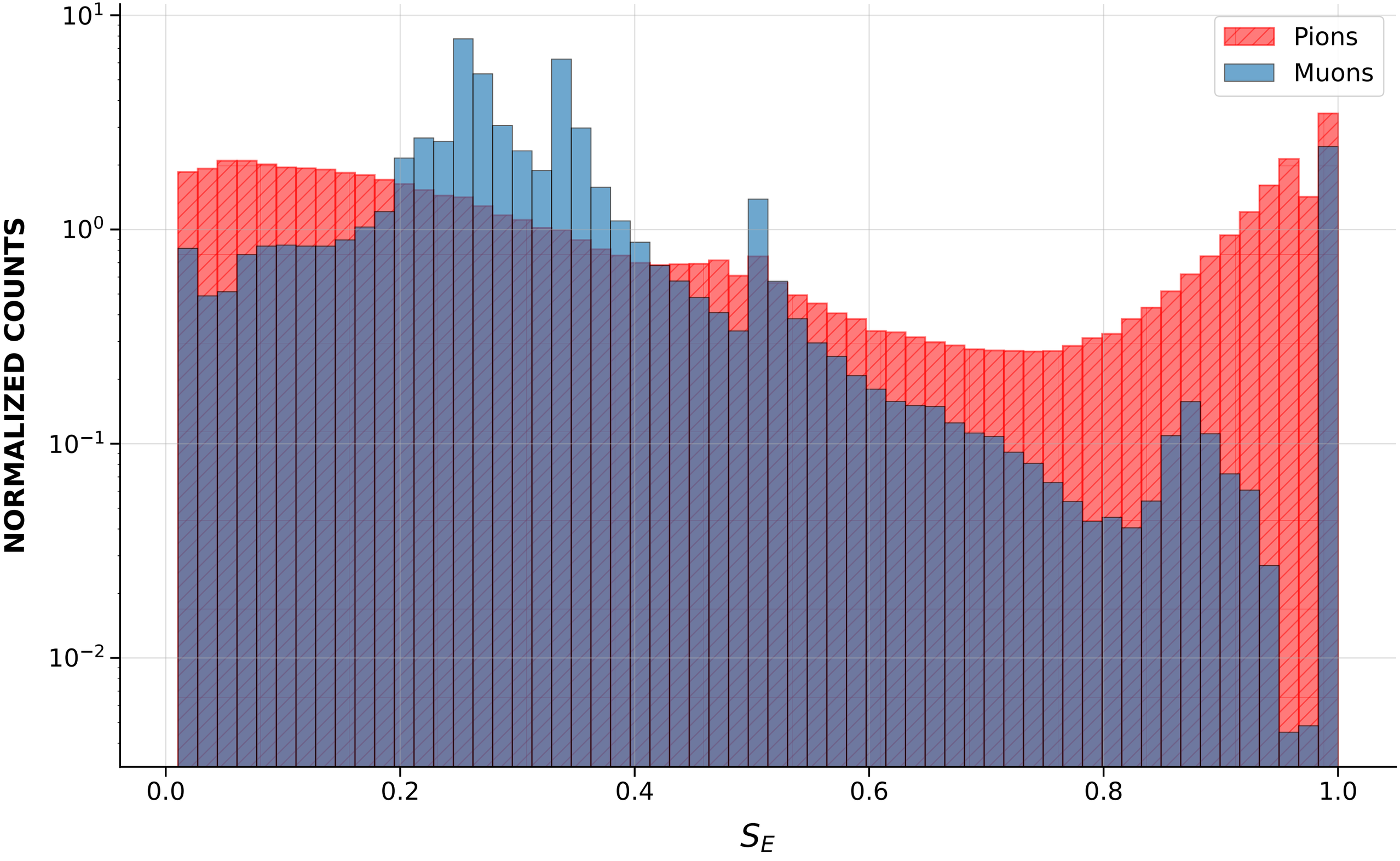
ENGINEERED CALORIMETER FEATURE
logHCal



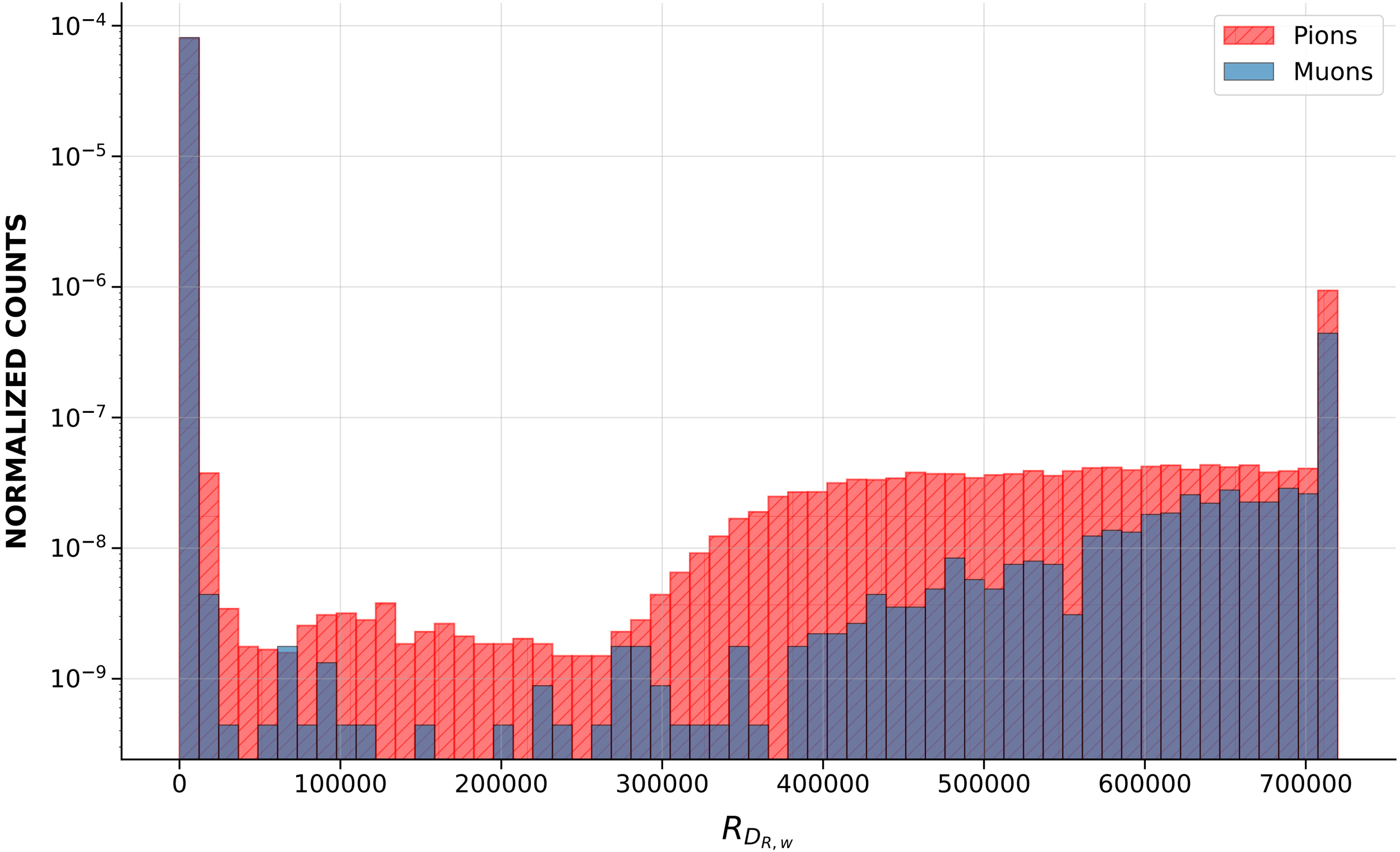
ENGINEERED CALORIMETER FEATURE
EnergyStdDev_ratio



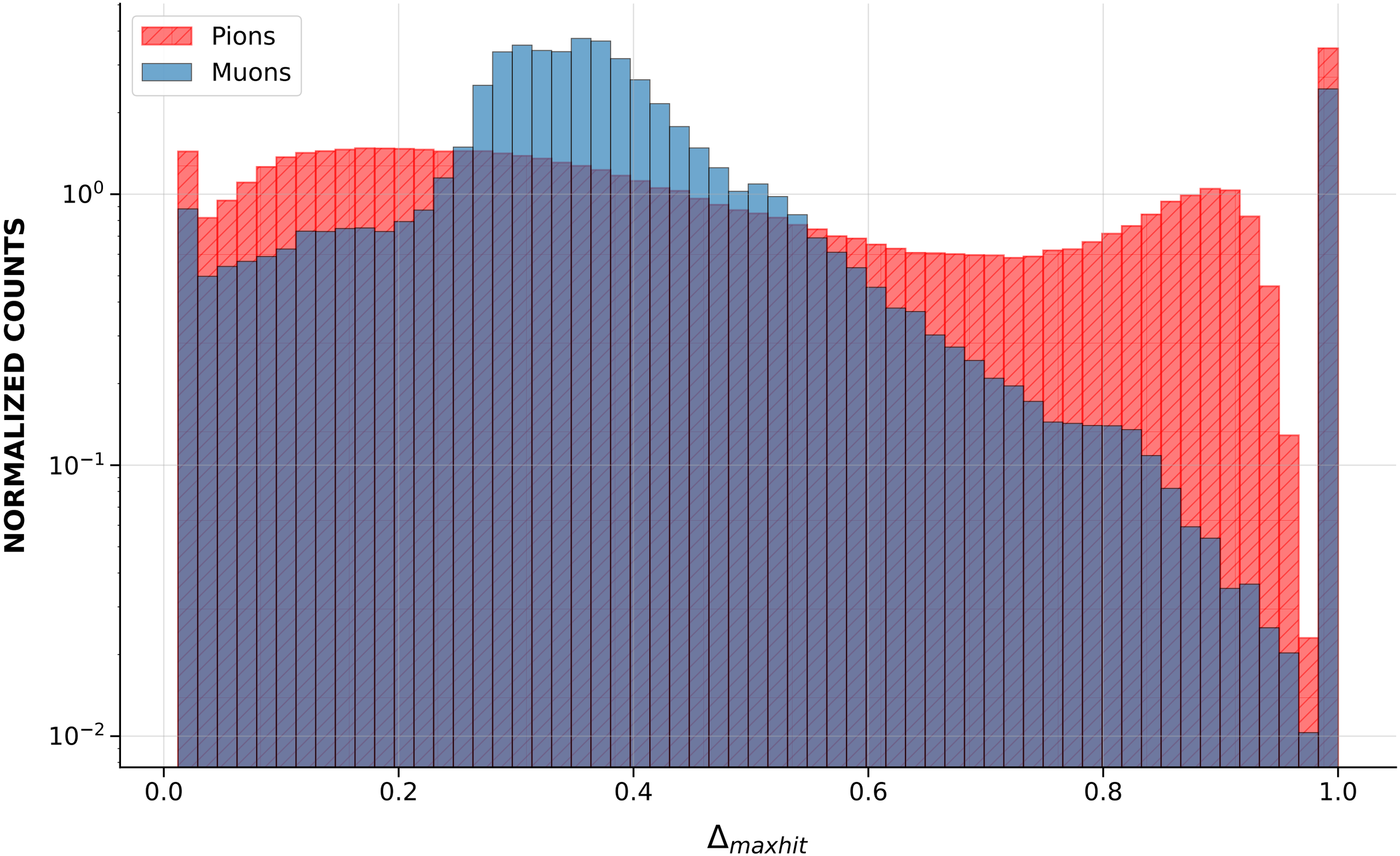
ENGINEERED CALORIMETER FEATURE
EnergyConcentration_mismatch



ENGINEERED CALORIMETER FEATURE
disp_ratio



ENGINEERED CALORIMETER FEATURE
MaxHitFrac_mismatch



ENGINEERED CALORIMETER FEATURE
HitRatio

